

TM. -02

What each YAML Control

problem.yaml:

It basically answers what computing / workload the accelerator has to run.

Change this file to evaluate a different layer (e.g., a different convolution or a fully-connected layer) on the same hardware.

arch.yaml:

This file describes the *hardware organization* — the hierarchical tree of storage levels (register files, buffers, DRAM) and compute units (MACs), including their capacities, number of instances, bandwidth, and other microarchitectural attributes

mapper.yaml:

it configures how the searching should be done: which optimization metric to target (e.g., energy-delay product), which search algorithm to use (the current file uses exhaustive), how many search threads/trials to run, and stopping/convergence criteria (we can define something like stop after 100 iterations if no improvement is observed)

Timeloop takes `problem.yaml` + `arch.yaml` to construct the legal mapspace (Fig. 2 below), then uses `mapper.yaml`'s settings to drive the search loop that evaluates candidate mappings with the cost model and picks the best one — outputting the performance/energy/area result

TM-03

What is Loop Nest:

It is a set of nested for loops (used to mimic multidimensional computation like a CNN layer)

From the paper:

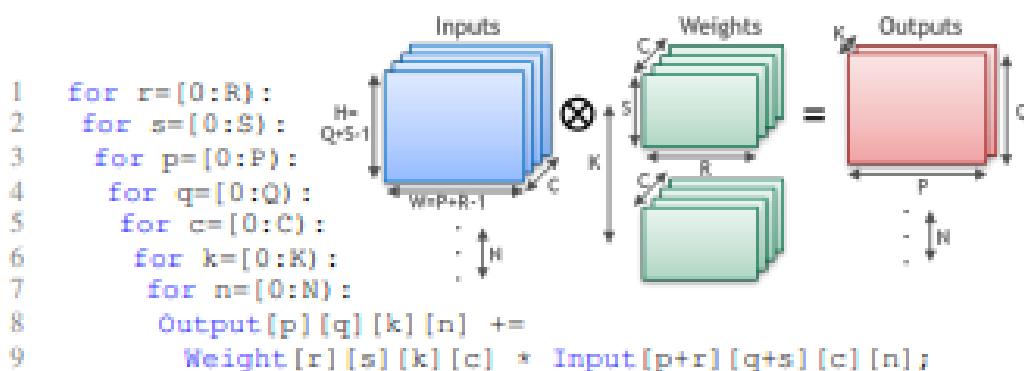


Fig. 3. Convolutional layer 7D loop nest.

The convolutional layer is written as 7 nested for loops, one for each of R, S, P, Q, C, K, N (filter height/width, output height/width, input channels, output channels, batch size).

Mapping and Mapper:

Mapping:

For a given workload run on a given architecture, a mapping is one specific choice of how to partition/order/schedule the computation and data movement. Different mappings on the *same* hardware can perform very differently.

Mapspace:

the set of all legal mappings for a workload on an architecture. Finding the best mapping means searching this space.

Mapper:

Its job is to find a good mapping for each design/workload pair by building the mapspace and searching it for the optimal mapping according to a given cost metric

Timeloop's model reports several statistics per mapping - performance (cycles/latency), energy.

Note:

Originally we were running only one instruction at a time and optimizing it, now with the current changes in the framework we are now running several operations sequentially and then optimize for this whole workload (tbd in Spatial Model Phase) which is the combination of these operations. This is consistent with the paper -

“To evaluate a complete network, one can invoke Timeloop sequentially on each layer and accumulate the results. Each layer has tremendous reuse opportunities, and we leave exploration of cross-layer reuse to future work.”

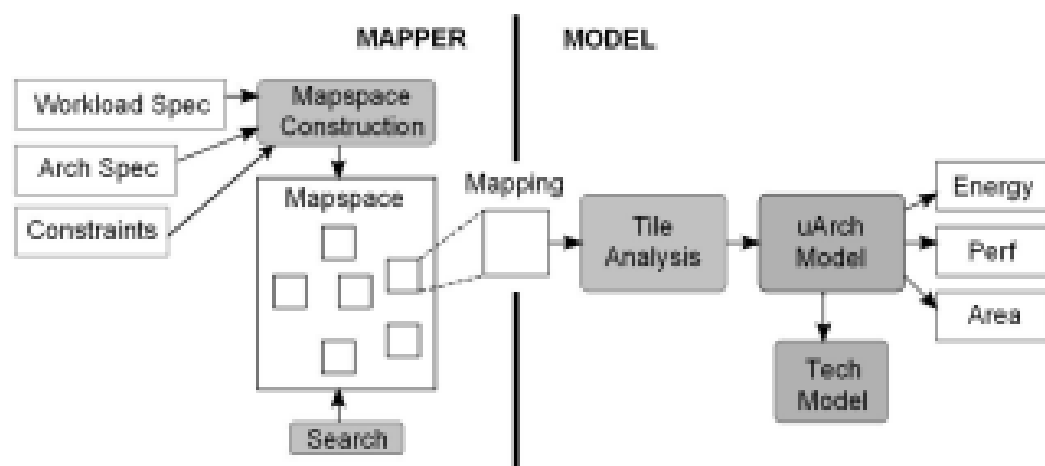


Fig. 2. Timeloop tool-flow.